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How I derived Quantum Mechanics from "Scratch" and Why it Matters / 🧵

They say that politics is downstream from culture, but very few appreciate the immense influence science has on culture too. Indeed you could say it even strongly: science defines culture, from the printing press, to 'enlightenment' and modern day scientism* which crippled the planet with a recent society-changing pandemic.

At the centre of science is the same question raised and answered by Aristotle: what is everything made of, what is the telos, or purpose of that substance? In ancient times all the way until a recent century, it was accepted that the elements were Earth, Fire, Water and Air, and the telos is for each of these elements to end up amongst each other. As the East, West Asia and most significantly, European men advanced science, this understanding was improved with better resolution. Chemistry, electricity, modern optics, astronomy and more.

At some point in time, academics partitioned science into two, the hard sciences and the soft sciences. The hard sciences are the true roots and so this is where my focus will lie.

The frontier of hard science is the intersection of quantum mechanics (QM) and gravitation. Quantum mechanics describes what things are made of and how they interact with each other, up to even macroscopic scales. Gravitation describes things at cosmic scales. We cannot reconcile our two theories together, and so at the large end we get contestable ideas like black holes and other nonsense which captures the imagination of society.

Darker still is what we get from our epistemology at the smaller scale: multiverses, intrinsic randomness, solipsistic observer dynamics and much much more. The current state of Quantum mechanics has captured the mind of anyone who is at least a little bit well read more than anything else in science. The influence has been nihilistic, a disconnected world that seems like a dream of one person. One idea that hit the mainstream is Quantum Immortality, promoted by the late John McAfee who participated in a game of Russian roulette in front of a journalist to demonstrate that the "timeline" was his and he couldn't die. Needless to say, this is absolute nonsense, but these sorts of belief influence people more than religion ever can due to the tangibility of science.

Most people don't get this influence directly however, it is always indirectly through people exposed to QM. For example, someone well versed in it might see the randomness as proof that there's no overall destiny or purpose to life. This leaks out into ardent support for mathematical impossibilities** like gradual/randomly driven Darwinian evolution rather than active adaptation. Another might see the observer effect, the idea that reality changes through any measurement or observation, as overturning the hierarchy of God's omniscience. They would then end up assigning it to themselves instead, in a Cartesian manner.

These people will go on to influence others indirectly, spreading atheism, nihilism and many ideas with full conviction. After all, in their perspective, the very fabric of reality supports their assertions. As the Hermes school says, as above so below and vice versa. The main problem is this: Those who know QM through coursework, or popular media, have learnt it entirely incorrectly. The pedagogy is completely wrong and I have proven it in my recently published draft Relativistic Soliton Mechanics (2025) [1]. In order to do so, I had to do something that was considered impossible. This thread will explain what that is, how it was proven, and the implications of this overturning of the pedagogical treatment of QM.

I'll go through just a few examples: Many leading physicists like Feynman claim that QM cannot be truly understood, and the famous Schrödinger's equation cannot be derived. I disprove this directly in the paper, by deriving it without "non-classical" postulation. All standard treatments claim an energy-frequency relation that is assigned to a mythical object known as a photon. I prove that it is a required relation for an electron, without postulation simply a process of elimination. The standard formulation of quantum mechanics requires one to throw away realism, the idea that the physical world exists objectively, independently of our minds or observations. I derive the same equations and relationships FROM realism.

The particular implications of that last result is staggering as it represents a return to ontology. As I am striking at the very fabric of what we understand reality to be, all the implications will naturally involve everything that lives on top of QM, from microbiology, biology, neuroscience, psychology, sociology, politics and more. The ontology spreads to everything (except of course, gravity -- for now? 😉).

The average person reading this probably doesn't know what QM even is, only that they heard it was "weird" and contained twisting facts within it, like the "particle-wave" duality or whatever is covered by high schools/YouTube today. This is a unique challenge for me to overcome, because simply explaining what QM is to the average person as it is written in textbooks, would take an entire set of courses, not a short Twitter thread. The majority won't get through it, as it requires a great deal of patience.

Thus, I will take a dual approach. This thread will be a qualitative description of my 104 page paper [1], extended to cover why it should interest those unfamiliar with this material.

This is going to be a long one, and I'm going to be writing it out live, so for your sake and mine, here is my planned outline of this thread:

Part A: Origin of QM through 200 years of scientific observations and why it was needed at all
Part B: What is wrong about how QM is taught and what we aim to fix. Measurements in intensity, frequency and energy.
Part C: Poincaré's tour de force, E=mc^2, inventing relativity, deriving electromagnetism from conservation laws and Lorentz invariance and his final word on matter being a field disturbance
Part D: Picking up where Poincaré left off with the most parsimonious possible derivation of non-interacting QM, relating matter's frequency and energy thus dispersion relation.
Part E: Connecting matter with the field, deriving the Schrödinger equation from scratch.

Part F: Reinterpretation of foundational experiments
Part G: Social implications and conclusion.

Let's begin with what QM is... or, let's say, why we needed to create it at all in the first place! Each part will be a reply to this thread so stay tuned as I write it out.

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* Often called "soyience".
** See the problem of irreducible complexity and the vigorous and often times nonsensical debate surrounding it, e.g. 2

Please note: If you are already familiar with QM, please skip this 'popular science' thread and read my paper directly: [1].

[1] https://doi.org/10.5281/zenodo.18109461
[2] https://scienceandculture.com/2011/03/michael_behe_hasnt_been_refute/

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Some Version of You Always Beats Death, According to This Scientific Theory

This idea puts a morbid twist on quantum theory, suggesting that a copy of you may remain alive, no matter what.

BY ELANA SPIVACK PUBLISHED: APR 22, 2025 2:51 PM EDT

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